energy is $p^2/2m$ and work out the Schrödinger equation (the original one that Schrödinger discovered) for a free particle.

Our plan is to follow the same process we used in the previous example, using the Hamiltonian to write a time-dependent Schrödinger equation. As usual, the left side of the equation is

$$i\hbar\frac{\partial\psi}{\partial t}.$$

We'll derive the right-hand side by rewriting the classical Hamiltonian—the kinetic energy—as an operator. The classical kinetic energy is

$$p^2/2m.$$

The quantum version replaces *p* with **P:**

$$\mathbf{H} = \mathbf{P}^2/2\mathrm{m}.$$

What is the meaning of this? As we've seen, the operator **P** is defined as

$$\mathbf{P} = -i\hbar\frac{\partial}{\partial x}.$$

The square of **P** is just the operator that you get by allowing **P** to act twice in succession. Thus,

$$\mathbf{P}^2 = (-i\hbar\frac{\partial}{\partial x})(-i\hbar\frac{\partial}{\partial x}),$$

or